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Third-order three-wave mixing in single-mode fibers: exact solutions and spatial instability effects

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Exact solutions are presented to the steady-state coupled-mode equations that govern the nonlinear parametric interaction of a central-frequency wave with a pair of upshifted and downshifted sidebands in isotropic single-mode optical fibers. This solution accounts for pump depletion as well for as a possible phase mismatch among the waves. The existence is predicted of eigensolutions propagating unchanged along the fiber, which may be either spatially stable or spatially unstable, depending on the total power and the propagation-constant mismatch. The presence of spatially unstable eigensolutions dramatically affects the power exchange among the three waves. The physical implications of this instability for the frequency-conversion process, as well as its potential application to all-optical switching, are discussed.

1. INTRODUCTION

Waves of different frequencies are mixed through optical nonlinearities to generate parametrically or to amplify new frequencies. For instance, the mixing of two waves through second-order nonlinearities [i.e., involving the susceptibility $\chi^{(2)}$] contributes to a third wave, giving rise to sum and difference frequency generation or to second-harmonic generation as a degenerate case. The nonlinear second-order parametric three-wave mixing in noncentrosymmetric bulk media was investigated in early studies of nonlinear optical interactions.^{1,2}

On the other hand, it is well known that parametric mixing of waves with different frequencies may also occur, even in media lacking inversion symmetry, through third-order nonlinearities.² While second-order nonlinearities involve the interaction of three photons, third-order susceptibilities involve the interaction of four photons, and therefore parametric processes in the latter case are commonly termed four-photon mixing effects. Although this terminology is formally correct, misunderstanding may arise from the fact that four-photon mixing, which generally involves the interaction of four physically distinct (i.e., at different frequencies) waves, includes as a partially degenerate case (in which the frequencies of two photons are equal) the interaction of only three distinct waves. In the completely nondegenerate case, known also as parametric four-wave mixing (FWM), two pump waves at different frequencies and a signal wave interact to exchange energy with, or create, a fourth wave, usually called the idler. In contrast, when the frequencies of the two pump waves are degenerate, energy conservation requires that two pump photons be absorbed, thereby creating a pair of Stokes (i.e., at lower frequency) and anti-Stokes (i.e., at higher frequency) photons, whose frequencies are symmetrically located with respect to the frequency of the pump. For the sake of clarity, this partially degenerate interaction, which is the subject of the present paper, is denoted henceforth as third-order three-wave mixing (TWM).

The first observation of third-order TWM was reported in liquids.³ However, despite the weak third-order non-

linearity of silicate glasses, optical fibers turn out to be the best candidate for materials in which efficient third-order mixing processes may be observed.⁴⁻¹⁶ For this reason we explicitly refer in the following discussion to propagation along a single-mode fiber, though our analysis holds with minor modifications for the interaction of plane waves propagating collinearly along isotropic bulk media.⁹ We further limit our treatment to the case of collinearly polarized waves. In this case phase matching is possible, either by operating with small frequency detunings around the zero chromatic dispersion wavelength⁷ or in the anomalous dispersion regime whenever the nonlinear refractive index contribution to the wave-vector-matching condition compensates for the effect of chromatic group-velocity dispersion (GVD).^{13,17} The growth of an initially small modulation owing to the latter mechanism is also known as modulational instability.¹³

Note that the importance of the partially degenerate case also arises from the fact that the nonlinear evolution of a modulated wave in a single-mode optical fiber can be described in this context as the parametric TWM of the carrier wave, at frequency ω_0 , with the first two contiguous modulational sidebands at frequencies $\omega_0 \pm \Omega$.¹² On the other hand, since nonlinear parametric mixing effects require fulfillment of the phase-matching condition, the generation of sidebands at frequencies $\omega_0 \pm n\Omega$ is usually neglected in the analysis,^{9,12} as a higher-order non-phase-matched process. Therefore new frequencies are no longer generated, and one is concerned only with the parametric exchange of energy among the carrier wave and the sidebands.

In general, TWM may be described in either the time or the frequency domain. In the time domain the evolution of an intense stationary central-frequency wave with a superimposed modulation is governed by one or more coupled partial differential (nonlinear Schrödinger-like) equations that account for the presence of both nonlinearity and GVD.^{17,18} Though this approach has the advantage of including an arbitrary number of interacting sidebands, the solution of this problem is a complicated task, which is usually approached by resorting to linearization of the

equations. Linearized solutions allow for predicting important results such as modulational instability but do not yield information about the propagation beyond the initial stages. Approximate results that account for pump depletion, when GVD is neglected, were also obtained in simple form.¹⁹ On the other hand, in the frequency domain the mixing of N waves is governed by a set of N ordinary differential coupled-mode equations for the steady-state slowly varying amplitudes of the different waves.^{1,9,12} Though the solution of these equations for small N turns out to be an easier task, in earlier studies TWM and FWM were approached in the undepleted-pump approximation.^{9,12} Only recently has there been a considerable effort to analyze the fully nonlinear problem. Exact solutions, accounting for the effect of pump depletion, were given, for example, to the parametric FWM (two pump waves, a signal wave, and an idler wave) interaction.^{15,16}

In this paper we report analytical solutions to the coupled-mode equations that describe the nonlinear three-wave interaction in the frequency domain, without resorting to the undepleted-pump approximation (i.e., the initial amplitudes of the waves may be arbitrary). As we show, a deeper insight into the property of the solutions is obtained by carrying out the spatial stability analysis of the nonlinear eigenmodes of the interaction. In fact, as a consequence of the existence of spatially unstable eigenmodes, small variations in certain input conditions lead to quite different spatial evolutions and therefore to dramatically different output redistributions of power among the waves.

So far, spatial instabilities have been predicted for the nonlinear interaction of two modes, copropagating at the same frequency along different guiding structures.²⁰ The interacting modes can be those of two parallel waveguides coupled through evanescent tail overlap,²¹⁻²⁴ two polarization modes in a straight low-birefringence fiber^{25,26} or in a periodically twisted highly birefringent fiber,²⁷ the lowest order modes of a planar waveguide,²⁸ a structure with intersecting waveguides,²⁹ or a planar grating waveguide.³⁰ The common origin of the spatial instability in these nonlinear couplers must be ascribed to bifurcations of specific linear eigenmodes of the guiding structure. Ever since these instabilities were experimentally observed, they have attracted much interest in view of their potential application for performing all-optical switching.³¹⁻³⁴

The present study demonstrates that spatial instability effects also arise, through the susceptibility $\chi^{(3)}$, when the nonlinear interaction of three modes propagating with different frequencies is considered. From the theoretical point of view the present result reinforces the conviction that bifurcations of nonlinear eigenmodes and spatial instability effects generally occur whenever the nonlinear nature of the interaction of propagating modes is fully taken into account. On the other hand, from the application point of view exact solutions to TWM permit one to answer important unresolved questions. These concern, for instance, the evolution of an initially small modulation in the strong-interaction regime and the choice of the most favorable launching conditions in upfrequency or downfrequency conversion experiments when only one of the sidebands is injected at the input and pump depletion is not negligible. Moreover, the discovered sensitivity to the launching conditions, owing to the present instability,

is potentially exploitable for conceiving new schemes for all-optical switching.

This paper is organized in the following way: In Section 2 we report the coupled-mode equations and then outline the solution method. In Section 3 we discuss the spatial stability properties of the eigensolutions. Section 4 is devoted to an overview of the consequences of the spatial instability effect in typical experimental situations.

2. COUPLED-MODE EQUATIONS

Let us consider the interaction of three stationary collinearly polarized monochromatic waves at frequencies ω_0 , $\omega_0 - \Omega$, and $\omega_0 + \Omega$, characterized by the electric field complex amplitudes E_0 , E_1 , and E_2 , respectively. The total transverse electric field propagating along a single-mode optical fiber may be written as

$$E(x, y, z) = [E_0(z)\exp(ik_0z) + E_1(z)\exp(ik_1z + i\Omega t) + E_2(z)\exp(ik_2z - i\Omega t)]f(x, y)\exp(-i\omega_0 t), \quad (1)$$

where $k_0 = k(\omega_0)$, $k_1 = k(\omega_0 - \Omega)$, and $k_2 = k(\omega_0 + \Omega)$ are the propagation constants evaluated at the relative frequencies and $f(x, y)$ is the common transverse modal profile. We assume that the three waves interact through the intensity-dependent refractive index (i.e., a purely Kerr-like nonlinearity) of the fiber, that the susceptibility is nearly constant over the range of frequencies considered, and that the Raman-scattering process does not affect the parametric exchange of energy among the waves (i.e., the susceptibility is real). We further consider that the process is nearly phase matched, so that the usual assumption of neglecting the generation of higher-order harmonics can be made. Under these hypotheses it is easy to show that the complex amplitudes of the three waves satisfy the following system of three coupled-mode equations (see Ref. 9 and Appendix B):

$$-i\frac{dE_0}{dz} = R[|E_0|^2 + 2(|E_1|^2 + |E_2|^2)]E_0 + 2RE_1E_2E_0^*\exp(+i\Delta kz), \quad (2a)$$

$$-i\frac{dE_1}{dz} = R[|E_1|^2 + 2(|E_2|^2 + |E_0|^2)]E_1 + RE_2^*E_0^2\exp(-i\Delta kz), \quad (2b)$$

$$-i\frac{dE_2}{dz} = R[|E_2|^2 + 2(|E_1|^2 + |E_0|^2)]E_2 + RE_1^*E_0^2\exp(-i\Delta kz), \quad (2c)$$

where $\Delta k = k_1 + k_2 - 2k_0$ is the low-power propagation-constant mismatch and we have normalized the modes so that $|E_i|^2$ directly gives the power in watts of the corresponding mode. Hence $R = 2\pi n_2/\lambda_0 A$, where n_2 is the nonlinear refractive index at frequency ω_0 , measured in square meters per watt (the total refractive index is $n = n_1 + n_2 I$, where I is the intensity), and A is the effective core area.⁹ Note that in the undepleted-pump regime (i.e., for $|E_1| \ll |E_0|$, $|E_2| \ll |E_0|$), and for a phase-matched interaction (i.e., $\Delta k = 0$), Eqs. (2) reduce to those considered in Ref. 12. However, Eqs. (2) also hold, in general, when the interaction is not phase matched, $\Delta k \neq 0$ accounting for both waveguide- and material-dispersive contributions. In Appendix B we discuss in detail how

Eqs. (2) for TWM are properly related to the equations that govern nondegenerate FWM.^{9,15,16}

The first terms on the right-hand-sides of Eqs. (2) are self-phase modulation terms, which determine self-induced power-dependent phase shifts, whereas the last terms are responsible for the energy exchange among the waves. The remaining terms determine a cross-phase interaction. Equations (2) are conveniently rewritten in terms of amplitudes and phases of the waves, say, $A_{0,1,2}(z) \equiv |E_{0,1,2}(z)|$ and $\phi_{0,1,2}(z)$, respectively, where $E_j = A_j \exp(i\phi_j)$, for $j = 0, 1, 2$. Since the total power

$$P_0 = A_0^2 + A_1^2 + A_2^2 \quad (3)$$

is conserved, as one can immediately verify from Eqs. (2), it is convenient to deal with normalized dimensionless amplitude variables. To this end, we introduce the normalized pump power $\eta(z) \equiv [A_0^2(z)]/P_0$ and the normalized sideband amplitudes $a_{1,2}(z) \equiv A_{1,2}(z)/P_0^{1/2}$ [from Eq. (3) we have that $\eta + a_1^2 + a_2^2 = 1$]. Finally, since the phases appear in the equations only through the combination $\phi(z) \equiv \Delta k z + \phi_1(z) + \phi_2(z) - 2\phi_0(z)$, the number of effective real equations may be reduced to the following four:

$$\frac{d\eta}{dz} = -4RP_0\eta a_1 a_2 \sin \phi, \quad (4a)$$

$$\frac{da_1}{dz} = RP_0\eta a_2 \sin \phi, \quad (4b)$$

$$\frac{da_2}{dz} = RP_0\eta a_1 \sin \phi, \quad (4c)$$

$$\begin{aligned} \frac{d\phi}{dz} = & \Delta k + RP_0[2\eta - (a_1^2 + a_2^2)] \\ & + RP_0\left[\eta\left(\frac{a_1}{a_2} + \frac{a_2}{a_1}\right) - 4a_1 a_2\right] \cos \phi. \end{aligned} \quad (4d)$$

It is worth noting that, as one can easily verify from Eqs. (4), the power flows from the wave at frequency ω_0 toward the sidebands or vice versa, depending on the sign of $\sin \phi$ and hence on the current value of $\phi(z)$. The maximum power transfer or gain is expected whenever $\phi = \pm\pi/2$, in perfect agreement with experimental results.¹² Moreover, the information about the phase matching⁹ is contained in Eq. (4d) for the phase. The first and second terms on the right-hand-side of Eq. (4d) are, respectively, the linear phase mismatch Δk and the contribution to the wave-vector mismatch, say, $\Delta k_{nl} = RP_0(2\eta - a_1^2 - a_2^2)$, due to the self- and cross-phase interaction terms. Phase matching near the zero-dispersion point may be obtained for small frequency detunings at relatively low power, so that $\Delta k \approx 0$ and $RP_0 \ll 1$ in Eq. (4d).⁷ Conversely, let us consider the propagation of two weak sidebands in the presence of an intense wave at frequency ω_0 , so that Δk_{nl} can no longer be neglected. Since $a_1^2 + a_2^2 \ll \eta \approx 1$, we obtain $\Delta k_{nl} \approx 2RP_0$. Whenever the nonlinear contribution compensates for the linear phase mismatch, so that $\Delta k = -\Delta k_{nl} = -2RP_0$, a generalized phase-matching condition is fulfilled.⁹ In this case the first two terms in the right-hand-side of Eq. (4d) vanish and the growth rate of the sidebands is at a maximum, provided that the initial phase is correctly

chosen. Note that once the linear propagation constants are expanded around the central frequency ω_0 , we have $\Delta k = \beta''\Omega^2$, where $\beta'' = d^2k/d\omega^2|_{\omega_0} = 2\pi cD/\lambda_0^2$ stands for GVD. Phase matching is accomplished for $\beta'' < 0$ (i.e., in the anomalous dispersion regime) for a detuning $\Omega = (2RP_0/|\beta''|)^{1/2}$. This also turns out to be the detuning that yields the maximum gain in the modulational instability picture.^{13,17} In contrast, when we consider a weak central-frequency wave propagating in the presence of two intense detuned waves, we have $\eta \ll a_1^2 + a_2^2 \approx 1$ and $\Delta k_{nl} \approx -RP_0$. In the normal dispersion regime³⁵ for a frequency detuning $\Omega = (RP_0/\beta'')^{1/2}$, the maximum power transfer from the intense detuned waves to the weaker one is also expected in this case when the phase-matching condition $\Delta k = RP_0$ is fulfilled.

The system of equations (4), besides conserving total power, admits of the following two additional invariants (i.e., quantities that are conserved at any propagation distance along the fiber):

$$\alpha = a_1^2 - a_2^2, \quad (5)$$

$$H = 4\eta a_1 a_2 \cos \phi - (\kappa - 1)\eta - \frac{3}{2}\eta^2, \quad (6)$$

where we introduce the normalized mismatch $\kappa \equiv \Delta k/RP_0$.

The invariance of α physically represents conservation of the power difference between the two sidebands [Eq. (5) is a Manley-Rowe relation^{1,36}], and it is related to the fact that energy exchange among the waves implies that two photons at frequency ω_0 are absorbed (or emitted), through which contributions are made to *both* sideband frequencies. Note that the invariance of α , though not explicitly stated, is also satisfied for the linearized system of Ref. 12. Furthermore, being TWM symmetric with respect to the interchange of the Stokes and anti-Stokes waves, the solutions depend only on the modulus of α . The invariant H also depends on the phase relation between the fields (in contrast, the expressions for P_0 and α contain only the field moduli). Invariants of this form, related to the conservation of the time-averaged energy of the electromagnetic field, have already been recognized to exist for other parametric interactions such as second-order TWM¹ and nondegenerate FWM.^{15,16}

The system of equations (4) may be reduced, by means of the three independent invariants, to a single self-consistent equation for one of the four variables, which is therefore integrable at least by quadrature.³⁷ Since we are concerned most with energy exchange among the waves, we rewrite Eq. (4a) for the normalized power $\eta(z)$ by expressing the sideband amplitudes as a function of η through the invariants P_0 and α , and of $\sin[\phi(z)]$ through H . We obtain

$$\frac{d\eta}{d\xi} = \pm 2 \left[\frac{7}{16} f(\eta) \right]^{1/2}, \quad (7)$$

with

$$\begin{aligned} f(x) = & x^4 - \frac{20}{7} \left(1 + \frac{3}{5} \kappa \right) x^3 + \frac{4}{7} [3 - 4\alpha^2 - 3H] \\ & - \kappa(\kappa - 2)x^2 + \frac{8}{7} H(1 - \kappa)x - \frac{4H^2}{7}, \end{aligned}$$

where we introduce the normalized propagation length $\xi \equiv zRP_0$ and the plus (minus) corresponds to a negative (positive) value of $\sin[\phi(z)]$, which in turn corresponds to power flowing from the sidebands (pump) toward the pump (sidebands). Equation (7) may be also interpreted as the evolution equation for the velocity of an ideal point moving in a potential well $V(\eta)$ with total conserved energy E . It may be integrated to yield the solution in implicit form as

$$\xi = \frac{\pm 1}{\gamma} \int_{\eta(\xi=0)}^{\eta(\xi)} \frac{1}{[f(\eta)]^{1/2}} d\eta = \frac{\pm 1}{\gamma} \int_{\eta(\xi=0)}^{\eta(\xi)} \frac{1}{\{2[E - V(\eta)]\}^{1/2}} d\eta, \quad (8)$$

where $\gamma = (7/4)^{1/2}$. The potential well, which explicitly reads as

$$V(\eta) = - \int \frac{\partial^2 \eta}{\partial \xi^2} d\eta = - \frac{1}{2} \left\{ \eta^4 - \frac{20}{7} \left(1 + \frac{3}{5} \kappa \right) \eta^3 + \frac{4}{7} [3 - 4\alpha^2 - 3H - \kappa(\kappa - 2)] \eta^2 + \frac{8}{7} H(1 - \kappa) \eta \right\}, \quad (9)$$

and the total conserved energy

$$E = \frac{1}{2} \dot{\eta}^2 + V(\eta) = -2H^2/7 \quad (10)$$

are uniquely specified by the initial conditions through H , α , and κ .

Equation (8) implicitly gives $\eta(z)$, which in turn furnishes $\alpha_1(z)$, $\alpha_2(z)$, and $\phi(z)$ through the three invariants. The integral in Eq. (8) is an elliptic integral and can be inverted in terms of periodic Jacobian elliptic or hyperbolic functions.³⁸ Different forms of the solution are possible, depending on the location of the roots of the fourth-order polynomial $f(\eta)$. Explicit expressions of the solution are reported in Appendix A.

Though the coupled-mode equations [see Eqs. (2)] are reduced to a single equation for the pump power, and hence are explicitly solved, a deeper insight into the property of the solutions can be gained only by investigating the existence and the stability properties of spatial eigen-solutions (or nonlinear eigenmodes, i.e., three waves with a certain phase relation and a relative power content that propagate unchanged along the fiber). Section 3 is dedicated to this analysis by introducing a Hamiltonian formalism. The reader who is interested in the subject of this paper mainly from the experimental point of view may jump to Section 4, where we discuss physical implications of the solutions.

3. HAMILTONIAN FORMULATION AND STABILITY ANALYSIS

For the purpose of investigating the existence of eigen-solutions and the dynamic behavior of the present interaction, it is convenient to express the system of equations (4) as a one-degree-of-freedom Hamiltonian system. Once the two amplitudes $\alpha_{1,2}$ are expressed as a function of the pump power η through the first two invariants [Eqs. (3)–(5)], we obtain from Eqs. (4) a system of two self-consistent coupled equations for the variables (η, ϕ) , which turn out to be conjugate through the one-dimensional conservative Hamiltonian

$$H = H(\eta, \phi) = 2\eta[(1 - \eta)^2 - \alpha^2]^{1/2} \cos \phi - (\kappa - 1)\eta - \frac{3}{2}\eta^2, \quad (11)$$

so that the system of Hamiltonian equations explicitly reads as

$$\frac{d\eta}{d\xi} = \frac{\partial H}{\partial \phi} = -2\eta[(1 - \eta)^2 - \alpha^2]^{1/2} \sin \phi, \quad (12a)$$

$$\frac{d\phi}{d\xi} = -\frac{\partial H}{\partial \eta} = (\kappa - 1) + 3\eta - 2\frac{1 + 2\eta^2 - 3\eta - \alpha^2}{[(1 - \eta)^2 - \alpha^2]^{1/2}} \cos \phi, \quad (12b)$$

where $\xi \equiv zRP_0$. The advantage of reducing the starting system of equations to a one-dimensional Hamiltonian system, besides being convenient from the formal point of view, is that it allows one immediate insight into the properties of the solutions, since they are related to the topology of the phase space. This may be done simply by drawing the level curves (i.e., defined by $H = \text{constant}$) in the phase space. Furthermore, the stability of equilibrium points may be readily investigated by recurring to a well-developed theory for Hamiltonian systems.³⁷

A. Equal Sideband Amplitudes

Let us discuss first the case of initially equal sideband amplitudes, which corresponds to setting $\alpha = 0$. Since the power difference between the Stokes and anti-Stokes waves is conserved, the amount of power exchanged (either gained or lost) with the wave at frequency ω_0 is equally split between the two waves, for any propagation length. It is of interest here to determine the spatial eigensolutions representing input conditions (i.e., the ratio of sideband power to pump power and the relative input phase, for given total power and wave-vector mismatch) that propagate unchanged along the fiber. These eigensolutions correspond in turn to the fixed points of the Hamiltonian, obtained by imposing vanishing derivatives in Eqs. (12), which in this case read as

$$\frac{d\eta}{d\xi} = \frac{\partial H}{\partial \phi} = -2\eta(1 - \eta) \sin \phi, \quad (13a)$$

$$\frac{d\phi}{d\xi} = -\frac{\partial H}{\partial \eta} = (\kappa - 1) + 3\eta - 2(1 - 2\eta) \cos \phi. \quad (13b)$$

From Eq. (13a) it follows that $\eta(\xi)$ remains constant along the fiber either for $\eta(\xi = 0) = 0$ and $\eta(\xi = 0) = 1$ or for different values of $\eta(\xi = 0) \equiv \eta_e$ that satisfy $\sin \phi(\xi = 0) = 0$ and therefore correspond to amplitude (i.e., $\phi = 0$) or frequency (i.e., $\phi = \pi$) modulation in the limit of small ratios of sideband power to pump power.

Whenever $\eta = 0$, the three-wave interaction reduces to the cross-phase interaction of two beams with frequency separation 2Ω in the absence of the wave at frequency ω_0 . Strictly speaking, $\eta = 0$ is an eigensolution only when the phase remains constant for any propagation distance. From Eq. (13b), with $d\phi/d\xi$ calculated for $\eta = 0$, it is readily verified that this occurs whenever $\phi = \cos^{-1}[(\kappa - 1)/2]$. When this relation is not satisfied, the phase of the two waves is no longer constant during propagation. However, since no power exchange occurs among the three waves even in this case, $\eta = 0$ may still be re-

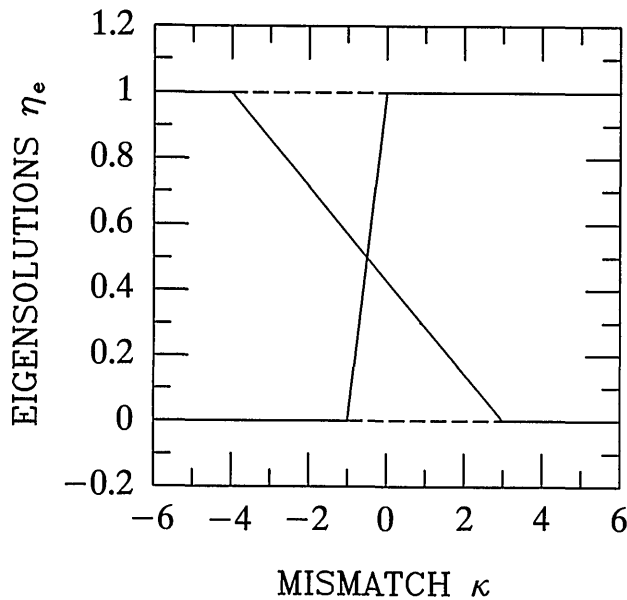


Fig. 1. Nonlinear eigensolutions for $\alpha = 0$: normalized pump power η_e for the stable (solid lines) and unstable (dashed lines) nonlinear eigenmodes versus normalized mismatch κ .

garded as an eigensolution of the TWM problem. Similar arguments hold for the solution $\eta = 1$, which physically corresponds to the propagation of the central-frequency wave in the absence of sidebands. In this case the proper phase of the eigensolution is found to be $\phi = \cos^{-1}[-(1 + \kappa/2)]$. Finally, the ratios of pump power to total power of the eigensolutions corresponding to amplitude (i.e., $\phi = 0$) or frequency (i.e., $\phi = \pi$) modulation are readily evaluated, under the restriction that $d\phi/d\xi = 0$, to be $\eta_e = (3 - \kappa)/7$ or $\eta_e = 1 + \kappa$, respectively.

In Fig. 1 we report the values of the normalized pump power $\eta = \eta_e$ corresponding to all the spatial eigensolutions versus the normalized mismatch κ . The dashed line indicates that the corresponding eigensolution is spatially unstable, while the solid lines denote stable eigensolutions. As Fig. 1 shows, $\eta_e = 0$ and $\eta_e = 1$ are the only stable eigensolutions for sufficiently large absolute values of normalized mismatch κ . As the value of κ decreases, being either positive or negative, these eigensolutions lose stability in correspondence to values of κ for which two branches of new stable eigensolutions appear. These correspond to the $\phi = 0$ or $\phi = \pi$ eigensolutions, for which the dependence of η_e on κ is linear. This exchange of stability between eigensolutions, which are linear functions of one parameter in the parameter space, is a standard characteristic of nonlinear dynamical systems that exhibit transcritical bifurcations.³⁹ In the case investigated in this paper the unstable branches of equilibrium points corresponding to $\eta = \eta_e > 1$ or $\eta = \eta_e < 0$, which would be present when two eigensolutions exchange their stability, do not assume physical significance.

It is worth noting that the eigensolutions $\eta_e = 0$ and $\eta_e = 1$ are unstable in a range of values of normalized mismatch κ centered on a value (i.e., $\kappa = 1$ for $\eta = 0$, and $\kappa = -2$ for $\eta = 1$) that corresponds to the nonlinear phase-matching condition discussed in Section 2.

The topology of the phase space is deeply affected by the existence of spatially unstable eigensolutions as shown in

Figs. 2–5, where we report the trajectories (or the level curves with constant Hamiltonian H) in the phase plane $(\eta \cos \phi, \eta \sin \phi)$ for different values of mismatch. Each closed trajectory line is the locus of points that represent the values of η and ϕ [a_1 and a_2 being attainable through the invariants of Eqs. (3), (5), and (6)] along the fiber for a given input condition, chosen from among those lying on the line. Note that the eigensolutions of Fig. 1 lie necessarily on the line $\eta \sin \phi = 0$, since they correspond to in-phase sidebands (amplitude modulation) if $\eta \cos \phi > 0$, or out-of-phase sidebands (frequency modulation) if $\eta \cos \phi < 0$. The eigensolutions $\eta_e = 0$ and $\eta_e = 1$ correspond in this phase plane to the origin and the unit circle, respectively.

In Fig. 2 we illustrate the topology of the phase plane for relatively high absolute values of the normalized mismatch ($\kappa = -4.5$ in this example). In these cases the only existing and stable eigensolutions are represented by the points $\eta = 0$ and $\eta = 1$ (see Fig. 1). As Fig. 2 shows, the

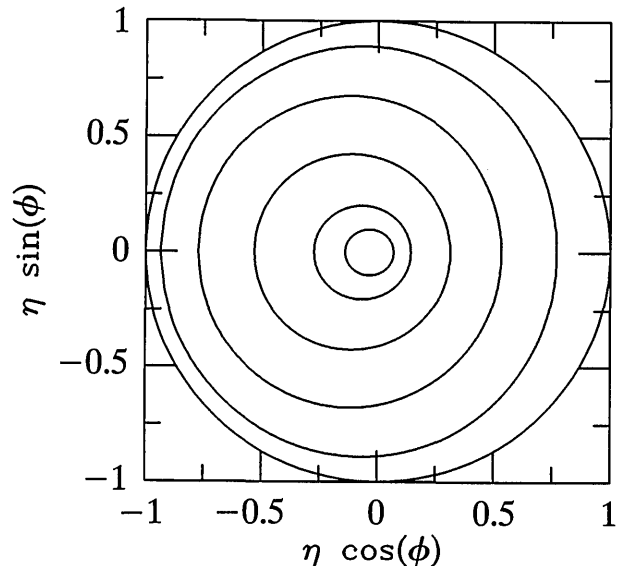


Fig. 2. Phase-space portrait for $\alpha = 0$ and $\kappa = -4.5$.

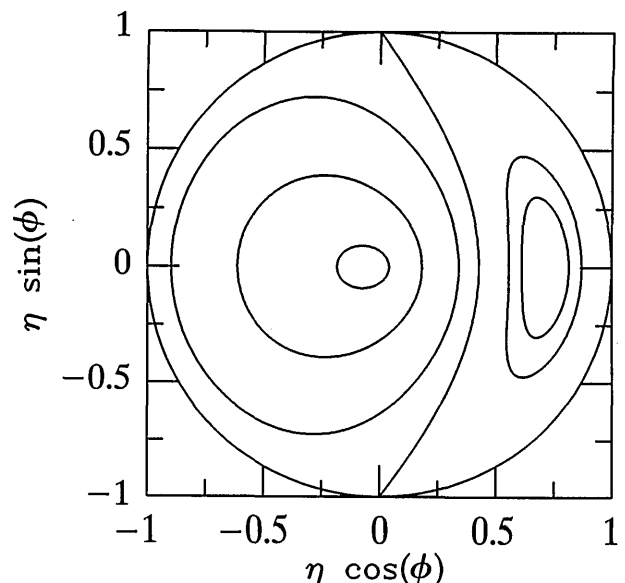
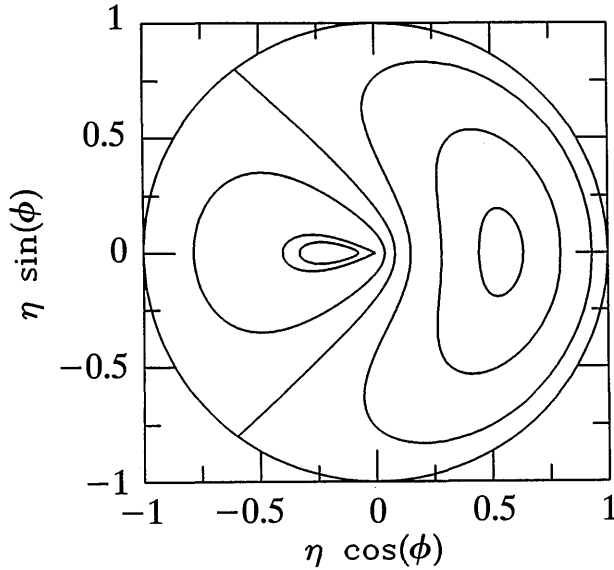
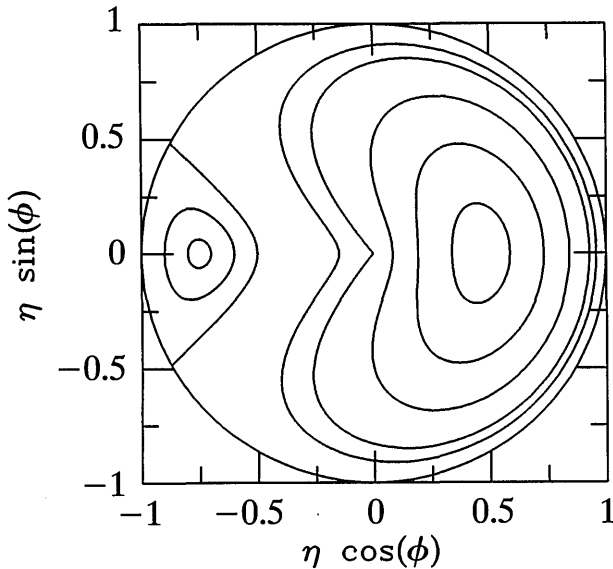


Fig. 3. Phase-space portrait for $\alpha = 0$ and $\kappa = -2$.

Fig. 4. Phase-space portrait for $\alpha = 0$ and $\kappa = -0.8$.Fig. 5. Phase-space portrait for $\alpha = 0$ and $\kappa = -0.25$.

trajectories are quite close to circumferences with constant η . Therefore only a moderate periodic power flow between the pump and the sidebands can be expected, in agreement with the fact that the interaction is highly mismatched. The amount of power exchanged among the waves decreases as the absolute value of κ grows larger.

The situation is rather different when we consider values of κ such that the eigensolutions $\eta_e = 0$ and $\eta_e = 1$ have exchanged their stability with new stable eigensolutions. In Fig. 3 we present the phase-space portrait that is attained whenever the interaction of a strong pump with weak sidebands is phase matched, so that $\kappa = -2$. We note that $\eta_e = 0$ is still a stable solution with surrounding orbits that correspond to periodic power flow among the waves, while $\eta_e = 1$ becomes unstable and a new stable eigensolution ($\eta = 0.72, \phi = 0$) appears. The trajectory that passes through the circle $\eta_e = 1$ is a separatrix trajectory, which divides two regions of periodic orbits, each surrounding one of the two stable elliptic points.

When κ is such that the eigensolution $\eta_e = 0$ loses its stability, the topology of the phase-space portrait changes radically. As Fig. 4 shows for $\kappa = -0.8$, whenever κ lies in the interval $[-1, -0.5]$, a new separatrix orbit appears, characterized by $H = 0$ and having a cusp in the origin (i.e., the hyperbolic point $\eta_e = 0$). This separatrix orbit grows larger as κ increases to $\kappa = -0.5$, where it merges with the separatrix orbit relative to $\eta_e = 1$. For this particular value of the mismatch the unstable eigensolutions $\eta_e = 0$ and $\eta_e = 1$ lie on the same orbit. When $-0.5 < \kappa < 0$, the two orbits that originated from the unstable points $\eta_e = 0$ and $\eta_e = 1$ are distinct again. They separate three distinct regions where periodic orbits exist, as Fig. 5 shows for $\kappa = -0.25$. Finally, when $0 < \kappa < 3$, the eigensolution $\eta_e = 1$ is stable, and only the separatrix $H = 0$, which has the shape of a cardioid, survives. The stability properties of the eigensolution $\eta_e = 0$, outlined above, may be clearly understood by examining the potential $V(\eta) - E$ as a function of η for $H = H(\eta = 0) = 0$ and different values of the mismatch κ (see Fig. 6). For $\kappa < -1$ or $\kappa > 3$ (see the dotted curve, for $\kappa = -2$), the potential $V(\eta)$ exhibits a minimum at $\eta = 0$, corresponding to stable equilibrium. Conversely, when $-1 < \kappa < 3$, the stability property of $\eta_e = 0$ changes, and the interaction is described by a motion taking place in a potential well with a local maximum at $\eta = 0$ (see the solid and dotted-dashed curves, for $\kappa = -0.5$ and $\kappa = 0$, respectively). For $\kappa = -0.5$ the potential exhibits another local maximum, at $\eta = 1$. In fact, in this case $H(\eta = 1) = 0$, and $\eta_e = 1$ is a spatial unstable eigensolution.

Finally, note that whenever the frequency detuning of the sidebands is such that $\kappa > 3$, the interaction is highly mismatched. The phase space looks like the one reported in Fig. 2 for $\kappa = -4.5$, and only moderate power exchange among the waves occurs.

B. Different Sideband Amplitudes

Let us now consider the case in which the launched sidebands no longer have equal power (i.e., $\alpha \neq 0$). This case

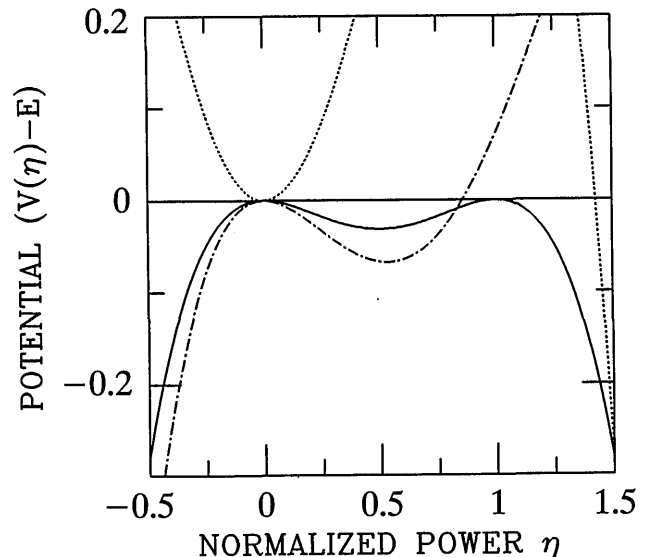


Fig. 6. Potential well $V(\eta) - E$ versus η for $\alpha = 0$, $H = 0$, and normalized mismatches $\kappa = -0.5$ (solid curve), $\kappa = 0$ (dotted-dashed curve), and $\kappa = -2$ (dotted curve). Physical limits require that $0 \leq \eta \leq 1$.

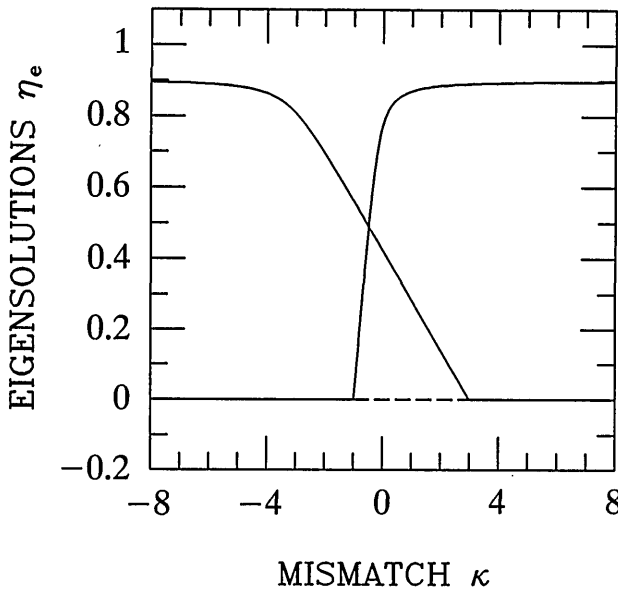


Fig. 7. Nonlinear eigensolutions for $\alpha = 0.9$: normalized pump power η_e for the stable (solid curve) and unstable (dashed curve) nonlinear eigenmodes versus normalized mismatch κ .

includes as a particular case the generation of the Stokes (anti-Stokes) wave when only the anti-Stokes (Stokes) wave is injected with the wave at frequency ω_0 .

For $\alpha \neq 0$, $\eta_e = 0$ is still a spatial eigensolution. In fact, no power exchange among the waves takes place when $\eta(z = 0) = 0$, while the two sidebands experience only an asymmetric power-dependent phase shift. It is worth noting, however, that though $d\eta/d\xi$ vanishes for $\eta = 1 - |\alpha|$, this is not an eigensolution. Whenever $\eta = 1 - |\alpha|$, one of the sidebands, say a_1 , vanishes. From Eqs. (4) it is evident that $d\eta/d\xi = da_2/d\xi = 0$, whereas $da_1/d\xi \neq 0$, implying that the fields E_0 and E_2 mix to generate parametrically the vanishing sideband E_1 .

As in the case discussed in Subsection 3.A, other eigensolutions $\eta_e \neq 0$ must have a phase relation ϕ such that $\sin \phi = 0$ (i.e., $\phi = 0$ or $\phi = \pi$). One may obtain the ratio of the central frequency to total power of these eigensolutions $\eta = \eta_e$ by imposing that $d\phi/d\xi = 0$, with $\phi = 0$ or $\phi = \pi$, respectively. As a consequence, it is straightforward to show that these eigensolutions may be found as the roots of the fourth-order polynomial

$$\begin{aligned} & -7\eta^4 + 6\eta^3(\kappa + 4) + \eta^2[\kappa(\kappa - 14) - 7(1 - \alpha^2) - 23] \\ & + 2\eta[3(1 - \alpha^2)(\kappa + 5) - (\kappa - 1)^2] + (1 - \alpha^2) \\ & \times [(\kappa - 1)^2 - 4(1 - \alpha^2)] = 0. \end{aligned} \quad (14)$$

The value of the normalized pump power η_e of the existing eigensolutions as a function of κ is summarized in Fig. 7 for $\alpha = 0.9$. The solution $\eta_e = 0$ is spatially unstable whenever $1 - 2(1 - \alpha^2)^{1/2} < \kappa < 1 + 2(1 - \alpha^2)^{1/2}$, whereas it exchanges stability with newly appearing eigensolutions for $\kappa = 1 \pm 2(1 - \alpha^2)^{1/2}$. In Fig. 7 these new branches of stable eigensolutions, which tend to $\eta = 1 - |\alpha|$ as $\kappa \rightarrow \infty$ or $\kappa \rightarrow -\infty$, correspond to the roots of Eq. (14) with $\phi = \pi$ or $\phi = 0$, respectively.

4. DISCUSSION OF RESULTS

The existence of spatially unstable eigensolutions has remarkable consequences for the dynamic behavior of

the nonlinear parametric interaction of three waves in a single-mode fiber. This section discusses the physical evolution of the three waves predicted on the basis of the theory outlined in the previous sections, in three cases that we believe to be relevant from the experimental point of view. In particular, we consider the case of symmetric sidebands (i.e., with the same amplitudes) with arbitrary relative phase, initially either much weaker or much stronger than the wave at frequency ω_0 , and the case of the parametric generation of a sideband when only the symmetric sideband is injected into the fiber.

A. Modulational Instability: Amplification of Small Symmetric Sidebands

Let us consider first the case of initially symmetric weak sidebands. This case obviously describes the propagation of a modulated intense carrier (for small modulation depths in the case of frequency modulation).

When pump depletion effects are neglected, as we discussed in Section 2, the maximum growth rate of the sideband is predicted when the nonlinear phase-matching condition is satisfied. This happens whenever $\kappa = -2$, or equivalently for a frequency detuning of the sidebands, $\Omega_{pm} = (2RP_0/|\beta''|)^{1/2}$. Provided that the detuning of the sidebands is fixed, the power flows from the pump to the sidebands if $\sin \phi_0 > 0$, the maximum gain being attained when $\phi_0 = \pi/2$.¹² However, an important issue, so far unaddressed, concerns the conversion efficiency in the strong-interaction regime (i.e., when pump depletion is not negligible). In fact, as pump depletion effects become nonnegligible, the frequency detuning of the sidebands that yields phase matching starts shifting toward lower values. This is because the power-dependent terms in the wave-vector-matching condition originating from the nonlinear refractive index [i.e., $\Delta k_n = RP_0(2\eta - a_1^2 - a_2^2) \approx 2RP_0\eta$] decrease as the pump depletes. Therefore the initially phase-matched interaction becomes progressively mismatched as the power is converted to the sidebands. When the phase-mismatch terms in Eqs. (4) become effective, the rate of power transfer saturates, which reduces the conversion efficiency. Conversely, let us consider a frequency detuning of the sidebands, $\Omega < \Omega_{pm}$. Though the waves are initially phase mismatched, the phase-matching detuning Ω_{pm} decreases owing to the pump depletion along the fiber, and as a result the interaction becomes progressively phase matched. For such detunings better efficiencies may be achieved, up to a detuning $\Omega \approx \Omega_{pm}/2$, which yields ideally complete conversion. These qualitative considerations are indeed supported by the results predicted on the basis of our theory and shown in Figs. 8 and 9. In Fig. 8 we plot the pump power versus the propagation length, predicted for an initially phase-matched small modulation (i.e., the ratio of sideband power to total power is $1 - \eta_0 = 0.01$, and the frequency detuning is $\Omega = \Omega_{pm}$), propagating in the anomalous dispersion regime for different initial values of the relative phase $\phi(z = 0) = \phi_0$. The length of the fiber in this example is three times the coherence length L_c of the process, say, $L_c = 2\pi/\Delta k$.⁹ This corresponds to a pump excitation factor of $RP_0L = (2\pi/|\kappa|) \times (L/L_c) = 3\pi$. The dotted line shows that for $\phi_0 = -\pi/2$ the small sideband power is converted asymptotically to the pump power [see Eq. (A7) below]. When the initial phase is reversed, the

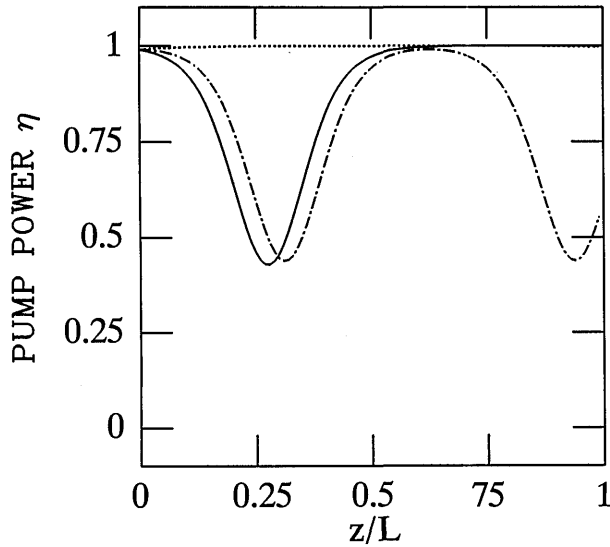


Fig. 8. Spatial evolution of the normalized pump power $\eta(z) = A_0^2/P_0$ along a fiber of length $L = 3L_c$ for an initially phase-matched (i.e., $\kappa = -2$, or $\Omega^2 = 2RP_0/|\beta''|$; and $\alpha = 0$) small modulation with $\eta_0 = \eta(z=0) = 0.99$ and initial phases $\phi_0 = \phi(z=0) = \pi/2$ (solid curve), $\phi_0 = -\pi/2$ (dotted line), and $\phi_0 = 0$ (dotted-dashed curve).

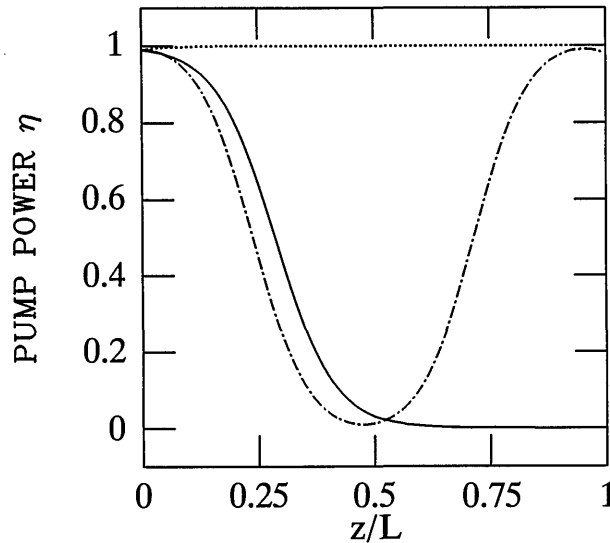


Fig. 9. Same as Fig. 8, except here along a fiber of length $L = 0.8L_c$ for an initially phase-mismatched (i.e., $\kappa = -0.5$, or equivalently $\Omega^2 = RP_0/2|\beta''|$; and $\alpha = 0$) small modulation with $\eta_0 = 0.99$ and initial phases $\phi_0 = 0.77\pi$ (solid curve), $\phi_0 = -0.77\pi$ (dotted line), and $\phi_0 = 0$ (dotted-dashed curve).

sidebands are initially amplified, then the power flow saturates and reverses, and the sidebands finally are depleted completely for $z = L$. This behavior corresponds to evolutions along the stable or unstable manifold of the separatrix originating from the unstable eigensolution $\eta_e = 1$ of the phase plane of Fig. 3. For any other value of the phase, such as $\phi_0 = 0$ (dotted-dashed curve), the power flows back and forth in a periodic fashion between the waves [see Eq. (A4) and evolutions along the periodic orbits of Fig. 3]. However, regardless of the initial relative phase, the conversion efficiency does not exceed 60%. The situation changes dramatically when the frequency detuning of the sidebands reduces to $\Omega \approx \Omega_{pm}/2$ (i.e.,

$\kappa \approx -0.5$), as Fig. 9 shows for a fiber of length $L = 0.8L_c$ (i.e., $RP_0L = 10$). The solid curve shows that for a proper initial phase $\phi_0 = \cos^{-1}[(\kappa - 1)/2] = 0.77\pi$, the pump tends to deplete asymptotically [see Eq. (A6)], converting all the power to the sidebands, whereas for $\phi_0 = -0.77\pi$ (dotted line) the sidebands deplete asymptotically. Different initial relative phases (see the dotted-dashed curve, for $\phi_0 = 0$) result in a periodic exchange of power. Even when the power exchange is periodic, almost complete energy exchange results. Asymptotic pump depletion to $\eta = 0$ similar to that shown by the solid curve in Fig. 9 [see Eq. (A6)] may take place for different initial conditions even if the condition $\eta \approx 1$ is not satisfied, provided that $-1 < \kappa < 3$, so that the eigensolution $\eta_e = 0$ is unstable. By imposing that the invariant H [see Eqs. (6) and (9)] calculated in the input state (η_0, ϕ_0) equal the invariant itself calculated in the output state (i.e., $\eta = 0$, which gives $H = 0$), we obtain the initial condition η_0 , which for given ϕ_0 and κ yields asymptotic pump depletion. These particular launching conditions (η_0, ϕ_0) are mapped in Fig. 10 for different values of the normalized mismatch. Since for $\kappa = -0.5$ the locus of initial conditions (η_0, ϕ_0) degenerates into a line, complete aperiodic pump depletion is predicted whenever the phase is $\phi_0 = 0.77\pi$, regardless of the initial modulation depth. Note also that for $\eta_0 < 0.85$ asymptotic pump depletion may take place even in the normal dispersion regime (i.e., for $\kappa > 0$).

Owing to the existence of spatially unstable eigensolutions, TWM may be useful for implementing new all-optical switching schemes. In these schemes the output state is discriminated by using a frequency-selecting optical element in order to isolate one of the waves, typically the pump. As we discuss below, switching may be induced either by small variations of the input modulation depth or relative phase, or by changes in the frequency detuning or the total power.

In Fig. 11 we show, as an example, the evolution of the pump power along the fiber for slightly different initial values of the sideband power (or modulation depth) while

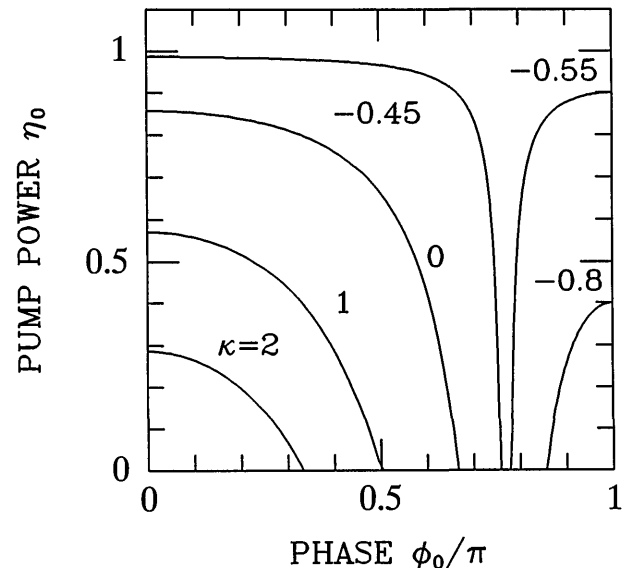


Fig. 10. Mapping of the initial conditions (η_0, ϕ_0) that evolve toward the unstable state $\eta_e = 0$ for different values of the normalized mismatch κ .

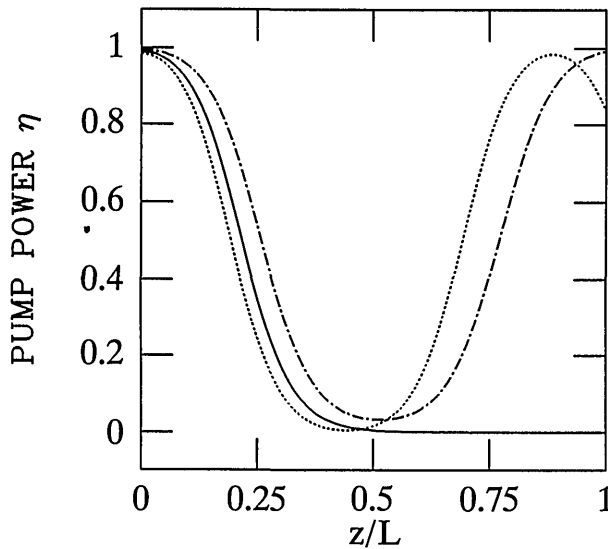


Fig. 11. Spatial evolution of the normalized pump power $\eta(z)$ along the fiber for symmetric sidebands (i.e., $\alpha = 0$), $\kappa = -0.465$, $\phi_0 = 0$, and slightly different initial ratios of pump power to total power: $\eta_0 = \eta(z=0) = 0.99$, $a_1^2 = a_2^2 = 0.005$ (solid curve); $\eta_0 = 0.995$, $a_1^2 = a_2^2 = 0.0025$ (dotted-dashed curve); $\eta_0 = 0.985$, $a_1^2 = a_2^2 = 0.0075$ (dotted curve). The fiber length is $L = L_c/2$.

the total power is maintained unchanged. The solid curve corresponds to the initial conditions reported in Fig. 10, chosen in such a way as to drive the interaction into the unstable state $\eta = 0$. For this case we chose the initial values $\phi_0 = 0$, $\eta_0 = 0.99$, $a_1^2 = a_2^2 = 0.005$, $\kappa = -0.465$, and fiber length $L = L_c/2$. For such launching conditions the pump depletes asymptotically to end up with a suppressed carrier. If the input sideband power is slightly increased or decreased in such a way as to maintain constant the total power (see the dotted and dotted-dashed curves, corresponding to $\eta_0 = 0.985$ and $\eta_0 = 0.995$, respectively), a periodic power flow results. Therefore dramatic variations of the output distribution of power may be controlled by small amplitude variations (in this example, $\Delta\eta_0/\eta_0$ corresponds to 0.5%) of weak-modulation sidebands. A peculiar characteristic of this coherent spatial instability is the possibility of controlling the relative contents of output power by also varying the relative input phase of the waves. Figure 12 shows how the transition from asymptotic pump depletion to periodic power exchange may be obtained by controlling the phase ϕ_0 . The input conditions correspond to a modulated beam with $\alpha = 0$, $\kappa = -0.15$, and $\eta_0 = 0.9$ propagating along a fiber whose length is $L = 0.5L_c$. Complete switching of the pump power is obtained by varying the phase from $\phi_0 = 0$ (solid curve) to $\phi_0 = \pi/2$ (dotted curve) or $\phi_0 = -\pi/2$ (dotted-dashed curve). Note that a negative value of $\sin \phi_0$ determines an initial power flow from the sidebands toward the pump. The flow is reversed at a value of η such that $V(\eta) = E$, corresponding to a turning point of the potential (see, for example, the dotted-dashed curve of Fig. 6).

So far, we have shown how the presence of spatially unstable solutions affects the evolutions of the three waves along the fiber when the launching conditions are slightly changed. All-optical switching may be induced by changing either the modulation depth or the relative

phase of the waves at the input. However, the evolution of a modulated beam with fixed modulation depth and relative phase may also be dramatically affected, as a consequence of the spatial instability, by small variations of total input power (or, equivalently, by frequency detuning). In Figs. 13 and 14 we show the computed transmissions, defined as the ratio of the output fraction to the input fraction of power at frequency ω_0 , say, $T = \eta(z=L)/\eta(z=0)$, versus the normalized total input power $p \equiv RP_0/|\Delta k| = 1/|\kappa|$ for different initial conditions and fiber lengths. Figures 13(a) and 13(b) correspond to a launched amplitude-modulated carrier (i.e., $\phi_0 = 0$) with $\eta_0 = 0.99$ and respective fiber lengths $L = 0.5L_c$ and $L = L_c$. Let us consider Fig. 13(a): when the power is so low that the nonlinearity is negligible, the waves do not exchange power, and $T = 1$. At relatively low power only a small amount of power flows among the waves, and for this fiber length the light initially present in the detuned waves is converted completely to the pump wave (i.e., $T > 1$). As the input power is increased, the amount of power exchanged among the waves grows larger, since the interaction is progressively phase matched. Meanwhile the period of the interaction decreases, since it depends inversely on the power [in Eq. (A5) below, the dependence on $(RP_0)^{-1}$ prevails for small k over the $K(k)$ dependence]. As a result the fraction of power coupled to the sidebands grows larger as the input power increases. At sufficiently high power (i.e., $p \approx 1.5$) almost complete power exchange takes place, while the period gets progressively larger [in Eq. (A5), $K(k)$ diverges logarithmically as k approaches 1^{38} , prevailing over the dependence on $(RP_0)^{-1}$]. At $p = 2.15$ the period becomes infinite and the pump depletes asymptotically [see the solid curve of Fig. 11 and Eq. (A6)], so that $T = 0$. As the power is further increased, the period decreases monotonically while the fraction of power exchanged progressively reduces. This causes secondary oscillations in the transmission curve, whose amplitudes reduce for large powers. As Fig. 13(b) shows for $L/L_c = 1$,

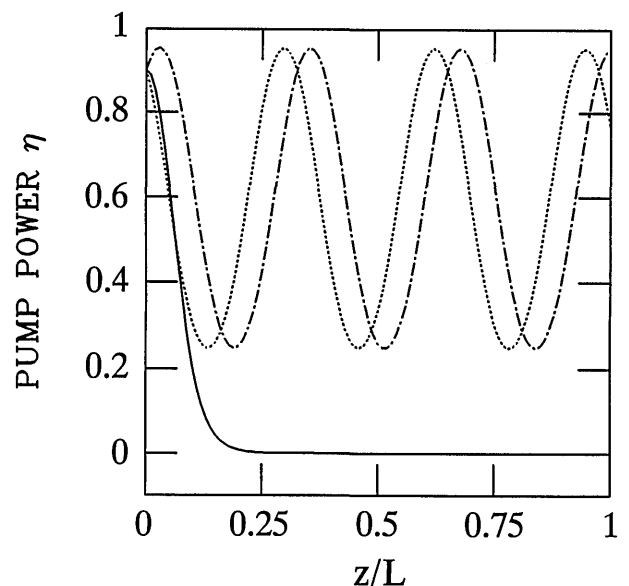


Fig. 12. Same as Fig. 11, except here for $\eta_0 = 0.9$, $\alpha = 0$, $\kappa = -0.15$, $L = 0.5L_c$, and initial phases $\phi_0 = 0$ (solid curve), $\phi_0 = \pi/2$ (dotted curve), and $\phi_0 = -\pi/2$ (dotted-dashed curve).

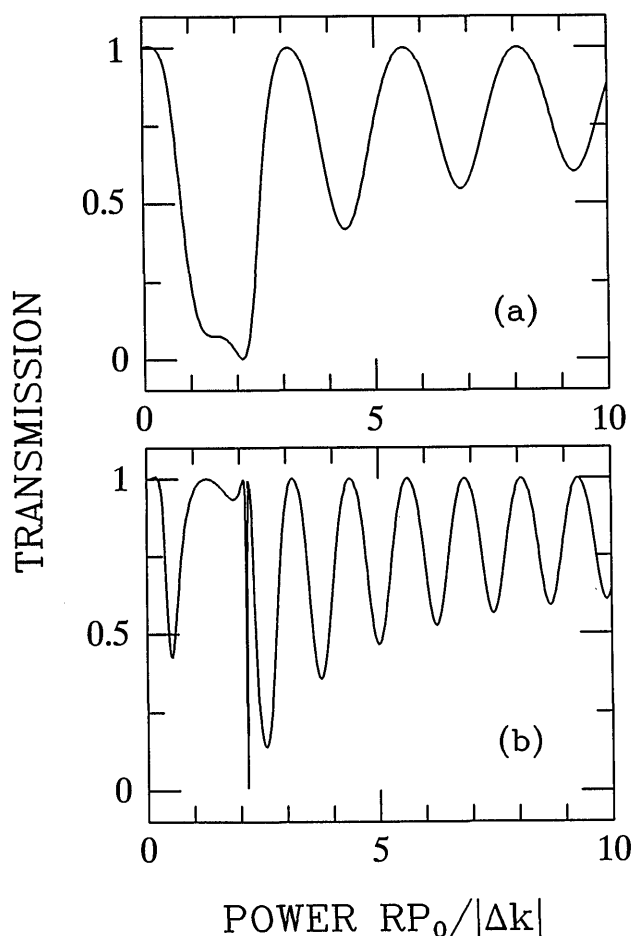


Fig. 13. Fraction of power, $A_0^2(z=L)/A_0^2(z=0)$, transmitted at frequency ω_0 in the anomalous dispersion regime as a function of the normalized total power $RP_0/|\Delta k| = 1/|\kappa|$ for $\eta_0 = 0.99$, $\phi_0 = 0$, and fiber lengths (a) $L = L_c/2$ and (b) $L = L_c$.

when the fiber length is increased, the transmission notch sharpens while the secondary bumps deepen.

In Figs. 14(a) and 14(b) we report for comparison the transmission when the launched beam is a purely frequency-modulated beam (i.e., $\phi_0 = \pi$) with $\eta_0 = 0.99$ and respective fiber lengths $L = 0.5L_c$ and $L = L_c$.

The behaviors shown in Figs. 13 and 14 should be readily observed in TWM experiments performed in a typical single-mode fiber. At an operating wavelength $\lambda \approx 1.3 \mu\text{m}$, with a dispersion $D = 2\pi c\beta''/\lambda^2 \approx -2.4 \text{ psec nm}^{-1} \text{ km}^{-1}$, the coherence length is $L_c = 2.8 \text{ km}$ for a modulation frequency $\Omega = 1 \text{ THz}$.¹³ Therefore, in a fiber of length $L = L_c/2 = 1.4 \text{ km}$, switching should be observed at a total power level $P_0 \approx |\beta''|\Omega^2/2R \approx 900 \text{ mW}$ (i.e., so that $p = RP_0L_c/2\pi \approx 0.5$), where typically $n_2 = 3.2 \times 10^{-16} \text{ cm}^2 \text{ W}^{-1}$ and $A = 1.3 \times 10^{-6} \text{ cm}^2$ ($R = 1.2 \times 10^{-3} \text{ W}^{-1} \text{ m}^{-1}$). Note that the power is inversely proportional to the coherence length, so that lower modulation frequencies result in lower powers, although the use of longer fibers is required.

Thus far, we have limited the discussion to an investigation of how the evolution of a modulated wave is affected by the existence of spatially unstable eigenmodes. However, we incidentally note that the existence of stable nonlinear eigenmodes may be relevant for the propagation of a modulated input in a telecommunication link. For

example, an investigation should be conducted of the possibility of counterbalancing the detrimental effect of dispersion, which would influence the evolution of a given input modulation during propagation, by means of the nonlinearity.

B. TWM with Initially Strong Stokes-anti-Stokes Waves

Let us now consider the opposite case, namely, that in which a pair of strong symmetric Stokes-anti-Stokes waves are launched along the fiber in the presence of a weak wave at frequency ω_0 .³⁵ This interaction may be useful for frequency conversion of two strong laser beams. As we discussed in Section 2, when the depletion of the strong waves is neglected, the maximum conversion toward the central-frequency wave is predicted for a fixed initial phase $\phi = -\pi/2$ whenever the interaction is phase matched. Phase matching occurs in this case in the normal dispersion regime for $\Delta k = -\Delta k_{nl} \approx RP_0$ (equivalently for $\kappa = 1$), where $P_0 \approx A_1^2 + A_2^2$. However, as the strong waves start to deplete, the power-dependent contribution to the propagation constants is responsible for a shift of the optimum frequency detuning $\Omega_{pm} = (RP_0/\beta'')^{1/2}$. As a consequence, an initially phase-matched interaction becomes progressively mismatched, and the power flow tends to decrease. This is clearly shown by the dotted curve of Fig. 15, representing the evolution of the wave at frequency ω_0 along a fiber of length $L =$

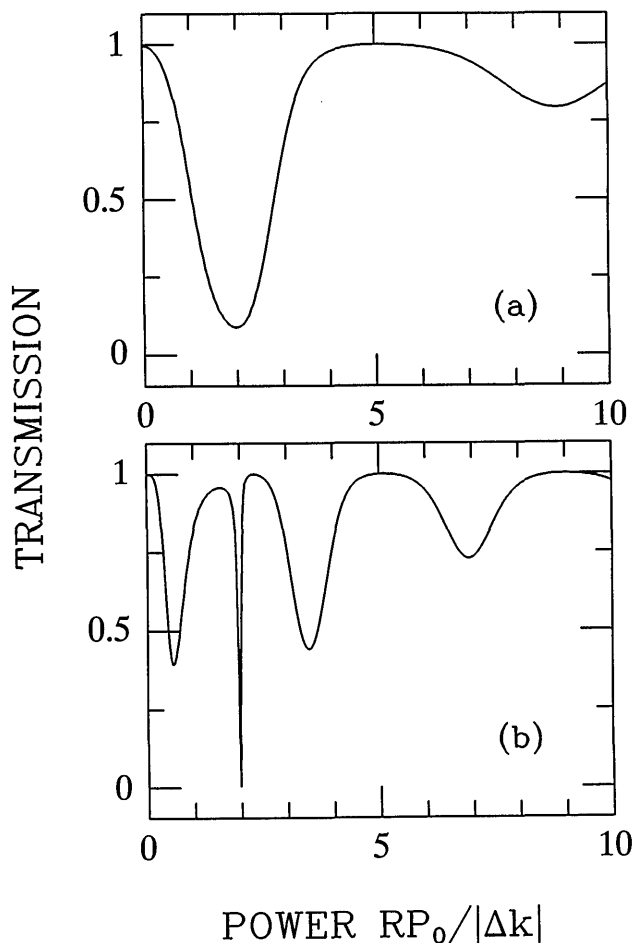


Fig. 14. Same as Fig. 13, except here for $\eta_0 = 0.99$, $\phi_0 = \pi$, and fiber lengths (a) $L = L_c/2$ and (b) $L = L_c$.

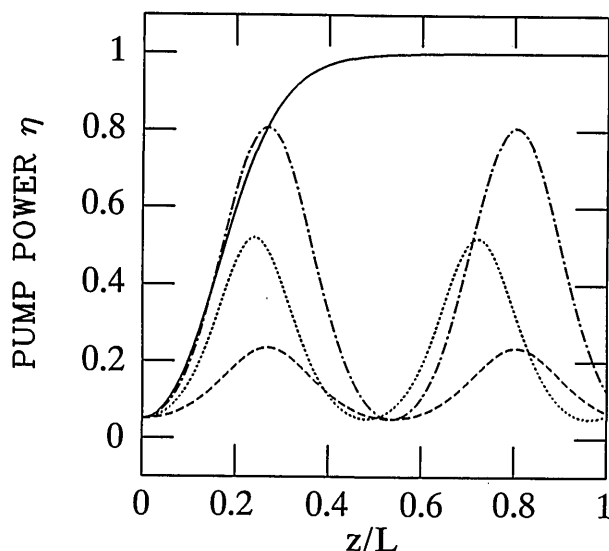


Fig. 15. Evolution of $\eta(z)$ along a fiber of length $L = 10/RP_0$ for initially strong sidebands $|a_1|^2 = |a_2|^2 = 0.475$, $\eta_0 = 0.05$, $\phi_0 = 0$, and normalized mismatches $\kappa = -0.675$ (solid curve), $\kappa = 0$ (dotted-dashed curve), $\kappa = 1$ (dotted curve), and $\kappa = 2$ (dashed curve).

$(5/\pi)L_c$ (corresponding to $RP_0L = 10$) when the launching conditions are such that $\eta_0 = 0.05$ and $a_1^2 = a_2^2 = 0.475$, and we set $\phi_0 = 0$. As shown by the dashed curve (for $\kappa = 2$), the situation is even worse for larger frequency detunings of the strong waves. In fact, pump depletion in this case contributes to an increase in the initial mismatch. Conversely, when pump depletion contributes to a progressive phase match of the interaction, larger conversion efficiency may be achieved. This is shown by the dotted-dashed and solid curves, corresponding to $\kappa = 0$ and $\kappa = -0.675$, respectively. In particular, the solid curve shows that for proper initial conditions ideal conversion occurs as the energy of the two strong waves is converted asymptotically to the wave at frequency ω_0 . This behavior corresponds to evolutions along a separatrix trajectory toward the unstable point $\eta = 1$ [see Eq. (A7) below]. This occurs whenever the initial conditions are chosen in such a way that the invariant H calculated at $z = 0$ is equal to $H(\eta = 1) = -(\kappa + 1/2)$. From Eq. (11) we obtain the optimum value of the normalized mismatch (and therefore practical values for the frequency detuning and the total power) for given launching conditions (η_0, ϕ_0) when $\kappa = [2\eta_0(1 - \eta_0)\cos\phi_0 + \eta_0 + 1/2(1 - 3\eta_0^2)]/(\eta_0 - 1)$. For $\eta_0 \ll 1$, the optimum conversion is obtained for $\kappa = -0.5$, corresponding to a frequency detuning $\Omega = (2RP_0/|\beta''|)^{1/2}$, in the anomalous dispersion regime. As we discussed in Subsection 4.A, this is also the optimum detuning, yielding ideal conversion for an input weakly modulated beam [i.e., for $a_1^2(z = 0) = a_2^2(z = 0) \ll 1$].

C. Frequency Conversion from a Single Sideband Wave

When a single sideband wave is injected at the input, the other sideband is parametrically generated through the interaction with the central-frequency wave. Therefore third-order TWM may be exploited for performing up-frequency, as well as downfrequency, conversion. In any frequency-conversion process it is obviously important to

determine the maximum efficiency that can be ideally obtained, as well as the launching conditions for reaching this ultimate efficiency. To this end, let us start by carrying out some qualitative considerations, which will then be supported by examples of exact solutions.

To simulate the generation of the new sideband from noise, we consider an initially strong pump $\eta_0 = 1 - \alpha - 2\varepsilon$, with a weak Stokes probe whose input power fraction is $a_1^2(z = 0) = \alpha + \varepsilon$. Here $\varepsilon = a_2^2(z = 0)$, which represents the noise level at the frequency of the anti-Stokes sideband, is much smaller than α , which in turn represents roughly the fraction of power of the probe. This is a special case of the TWM interaction with asymmetric sidebands, described by Eqs. (12) above, whose stability analysis was reported in Subsection 3.B. Since the sideband a_2 is generated from noise with the constraint that the power difference between the two detuned waves is conserved along the propagation direction [see Eq. (5)], the power converted from the pump is equally split between the Stokes and anti-Stokes waves. This implies that even if the pump can be entirely depleted, the maximum amount of power converted to the generated frequency is $A_2^2 = (1 - \alpha)P_0/2$, while the injected sideband is amplified to the power $A_1^2 = (1 + \alpha)P_0/2$. Note also that the pump cannot receive a net gain of energy from the injected sideband for values $\eta > 1 - |\alpha|$, since this requires absorption of photon pairs from both of the sidebands. Therefore the input value $\eta = 1 - |\alpha|$ is an upper limit that cannot be exceeded by the normalized pump power during propagation along the fiber.

One may think that high conversion efficiencies close to the ideal discussed limit could be obtained when the interaction is phase matched. As we discussed in Section 2, phase matching occurs whenever the nonlinear contributions to the propagation constants due to the cross-phase interaction compensate for the linear mismatch due to chromatic GVD. If the injected sideband is weak in comparison with the pump, the phase-matching condition becomes $\kappa \approx -2(1 - \alpha)$ in normalized units, which correctly reduces to $\kappa = -2$ for vanishing α . However, as we extensively discussed for the amplification of symmetric sidebands, as the pump starts to deplete, the frequency detuning, which yields phase matching, shifts inducing a saturation of the power flow among the waves. In Figure 16 we show the evolutions along a fiber of length $L = 4L_c$ of the pump and the generated sideband powers, $\eta(z) = A_0^2/P_0$ and $a_2^2(z) = A_2^2/P_0$, respectively. In this example the injected sideband is ten times weaker than the pump (i.e., $\eta_0 \approx 1 - \alpha = 0.9$, $a_1^2(z = 0) \approx \alpha = 0.1$), and the noise level is $\varepsilon = a_2^2(z = 0) = 10^{-5}$. We also set $\phi_0 = \pi/2$ and $\kappa = -1.8$, so that the interaction is phase matched. Though the maximum power that can be converted to the generated frequency is, in the present case, $A_2^2 = 0.45P_0$, the power is periodically exchanged among the waves, and the maximum conversion efficiency remains limited to less than 30%. Conversely, when the frequency detuning of the injected sideband is such that the interaction is initially phase mismatched, becoming progressively phase matched as the pump starts to deplete, a larger conversion efficiency may be achieved. In particular, since the point $\eta_e = 0$ is an unstable eigensolution even for $\alpha \neq 0$, as discussed in Subsection 3.B, we predict the existence of solutions where, for a proper de-

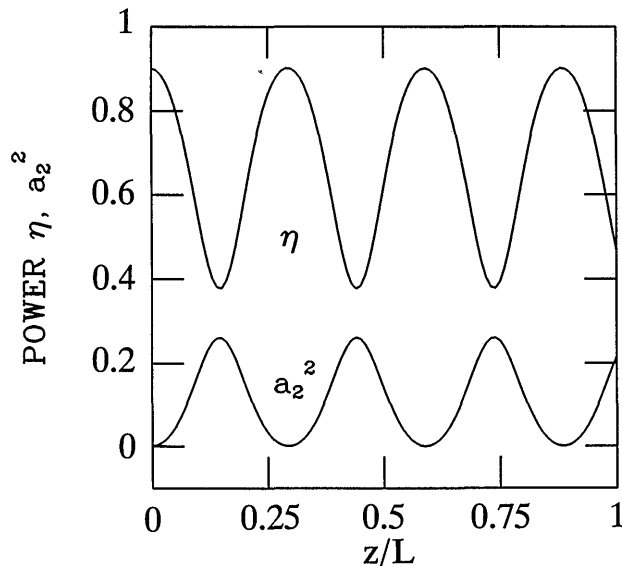


Fig. 16. Parametric generation of one sideband: spatial evolutions of the normalized pump and sideband powers, $\eta(z)$ and $a_2^2(z)$, respectively, for a phase-matched interaction with $\kappa = -1.8$, $\phi_0 = 0$, $\eta_0 \approx 1 - \alpha = 0.9$, $a_1^2(z=0) \approx \alpha = 0.1$, and $a_2^2(z=0) = \epsilon = 10^{-5}$. The fiber length is $L = 4L_c$.

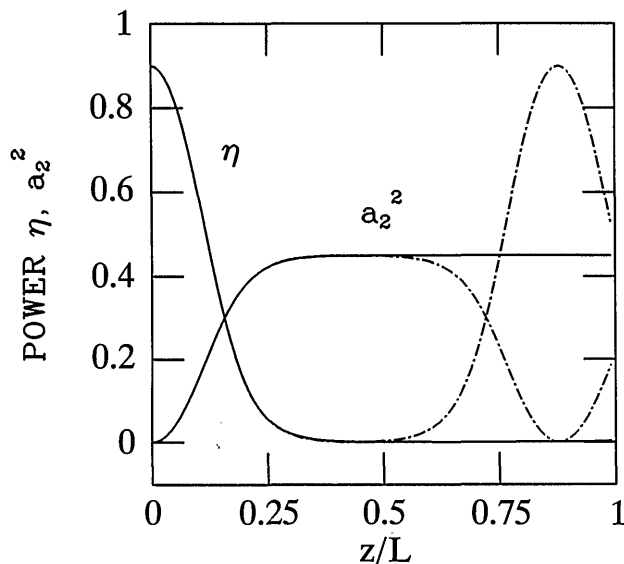


Fig. 17. Same as Fig. 16, except here for an initially phase-mismatched interaction with $\kappa = -0.35$ and initial phases $\phi_0 = \pi/2$ (solid curve) and $\phi_0 = 0$ (dotted-dashed curve). The fiber length is $L = 0.78L_c$.

tuning of the injected sideband, the pump tends to deplete asymptotically along the fiber, and the frequency generation efficiency reaches the maximum ideal value. These solutions are obtained for a value of the normalized mismatch κ (or equivalently for given frequency detuning and total power) that is a function of the probe fraction of total power α . In fact, by imposing that the invariant H [see Eq. (11) above] calculated in the input state $\eta \approx 1 - |\alpha|$ equal the invariant itself calculated in the output state (i.e., $\eta = 0$, which gives $H = 0$), we obtain the optimum value of normalized mismatch, yielding the maximum conversion efficiency as a function of α in the form $\kappa = -\frac{1}{2}f(\alpha)$, where $f(\alpha) = (3\alpha^2 + 1 - 4\alpha)/(1 - \alpha)$. Since

$f(\alpha = 0) = 1$, this condition reduces for $\alpha = 0$ to $\kappa = -0.5$, in agreement with what we found in Subsection 4.A for symmetric sidebands.

In Fig. 17 we report, as an example, the evolution of the pump and the generated sideband power along a fiber of length $L = 0.78L_c$ when a strong pump is injected with a weak sideband with optimum detuning (i.e., $\alpha = 0.1$, corresponding to $\kappa = -0.35$). The solid and dotted-dashed curves correspond to initial phases $\phi_0 = \pi/2$ and $\phi_0 = 0$, respectively. The solid curve shows that for a proper value of the relative phase the pump tends to deplete completely and the ideal conversion of 45% may be reached asymptotically as the waves propagate through the fiber. Other values of phases give rise to periodic exchange of power among the pump and the pair of injected and generated waves. Even in this case conversion efficiencies close to the maximum are reached periodically.

5. CONCLUSIONS

In this paper we have presented exact solutions to the steady-state coupled-mode equations that govern the nonlinear interaction of an intense wave with a pair of Stokes and anti-Stokes sidebands, propagating along a single-mode fiber. The existence of unstable eigensolutions has important consequences for the evolution of the waves and the power exchange. In particular, we showed that the notion of nonlinear phase matching, valid in the undepleted-pump regime, loses its validity as a criterion for achieving maximum conversion in the strong-interaction regime. Our solutions permit one to predict high efficiencies in the frequency-conversion process for initially mismatched waves, in every case of practical interest such as the amplification of a small modulation, the conversion from a pair of strong Stokes-anti-Stokes waves, and the conversion from an intense pump wave and one vanishing sideband. We believe that this conclusion holds in general for parametric processes of any order in the expansion of the nonlinear polarization, including completely degenerate cases of harmonic generation.

We further discussed how the present instability can be exploited for conceiving new schemes of all-optical switching.

Note that the effects predicted in this study could be hampered by the stimulated Brillouin scattering effect, which is well known to have a lower input power threshold.⁴⁰ However, the moderate nonlinearities of silica fibers require the use of pulses to observe nonlinear parametric mixing. This would be advantageous from one point of view, since it allows for suppression of stimulated Brillouin scattering,⁴⁰ although what remains is to investigate the actual possibility of observing the TWM instability in a pulsed regime. With reference to this point, we note that spatial instabilities for the two-mode interaction were first predicted on the basis of solutions to the stationary coupled-mode equations and then observed experimentally with the help of pulsed lasers. Only recently was it realized that the use of solitonlike pulses in the anomalous dispersion regime may greatly enhance the performance of all-optical switches based on the nonlinear interaction of propagating modes.^{41,42} This encourages us to extend the investigation of the present interaction to the nonstationary regime, eventually considering the three-

wave interaction of solitons in the anomalous dispersion regime. This will be the object of future investigations.

APPENDIX A

The usual procedure for solving an arbitrary system of coupled-mode equations that govern a steady-state nonlinear interaction consists, as shown in Section 2 for the present case, in reducing the original system to a single real equation for either the modal amplitude or phase, by means of a consistent number of conserved quantities. This equation may then be integrated by quadrature [see Eq. (8) above]. Whenever the quadrature integral contains a rational function of the square root of a quartic polynomial of the integral variable, the solution may be expressed explicitly in terms of the so-called Jacobian elliptic functions (which include as particular cases hyperbolic and circular trigonometric functions) by inverting the integral.³⁸ Solutions of this form are common for nonlinear optical interactions, including as examples second-order parametric mixing¹ and the coherent interaction of two modes in the different implementations of a nonlinear coupler.²⁰⁻²⁸ In this appendix we report explicit solutions for the pump power evolution for the present interaction [see Eqs. (8)-(10)].

The explicit form of the solution depends on the roots η_i ($i = 1, 2, 3, 4$) of the fourth-order polynomial in Eqs. (7) and (8):

$$f(\eta) = 2[E - V(\eta)] = (\eta - \eta_1)(\eta - \eta_2)(\eta - \eta_3)(\eta - \eta_4). \quad (\text{A1})$$

Once the initial conditions are fixed, the invariants (α, H) are specified and the coefficients of the polynomial $f(\eta)$ are uniquely determined through the values of κ , α , and H . The values of the roots then determine the specific expression of the solution. Here we report in detail both the cases of symmetric sidebands (i.e., $\alpha = 0$) and of optimum conversion with complete pump depletion from asymmetric sidebands (i.e., $\alpha \neq 0$, with $H = 0$). In these cases, if we rewrite $f(\eta)$ in a convenient way, it is clear that the four roots η_i of the equation

$$f(\eta) = \frac{1}{4}\{4\eta^2[(1 - \eta)^2 - \alpha^2] - [H + (\kappa - 1)\eta + \frac{3}{2}\eta^2]^2\} = 0 \quad (\text{A2})$$

assume particularly simple expressions, since Eq. (A2) reduces to two second-order algebraic equations. In the general case the solutions given below conserve their validity, whereas the root expressions become cumbersome functions of α , H , and κ .

1. Case I: Initially Equal Sideband Powers ($\alpha = 0$)

In the case of initially equal sideband powers the roots of Eq. (A2) are

$$\begin{aligned} \eta_{1,2} &= \frac{1}{4}\{-(\kappa - 3) \pm [(\kappa - 3)^2 - 14H]^{1/2}\}, \\ \eta_{3,4} &= \kappa + 1 \pm [(\kappa + 1)^2 + 2H]^{1/2}, \end{aligned} \quad (\text{A3})$$

where the first and the second subscript in each equation corresponds to the root with the positive and the negative sign, respectively. Note that whenever the initial conditions coincide with one of the stable eigensolutions re-

ported in Section 3, two of the roots in Eqs. (A3) coincide. This occurs either for $(\eta_0 = \eta_e = 1 + \kappa, \phi_0 = \pi)$, since $H = -(1 + \kappa)^2/2$, or for $(\eta_0 = \eta_e = (3 - \kappa)/7, \phi_0 = 0)$, since $H = (\kappa - 3)^2/14$. In these cases two of the roots coincide with the initial value $\eta = \eta_e$, which is a minimum of the potential $V(\eta)$; hence $\eta(\xi) = \eta_0 = \eta_e$. For any other choice of initial conditions such that $H \neq 0$ and $H \neq H(\eta = 1)$, the four roots are real and distinct. In this case, if we denote the roots by a, b, c , and d [ordered so that $a > b > \eta(z) \geq c > d$], the normalized pump power as a function of ξ becomes³⁸

$$\eta(\xi) = \frac{(b - c)\alpha \operatorname{sn}^2[(\pm\gamma\xi + \delta)/g] - (a - c)b}{(b - c)\operatorname{sn}^2[(\pm\gamma\xi + \delta)/g] - (a - c)}, \quad (\text{A4})$$

where sn stands for the elliptic sine function, $\gamma = (7/4)^{1/2}$, and the upper or lower sign must be chosen according to Eqs. (7) and (8). We also set

$$g = \frac{2}{[(a - c)(b - d)]^{1/2}}, \quad k^2 = \frac{(a - b)(a - d)}{(a - c)(b - d)},$$

$$\delta = \int_{\eta_0}^b \frac{d\eta'}{[f(\eta')]^{1/2}}.$$

Equation (A4) describes periodic power exchange with spatial period (in the physical coordinate z)

$$z_p \equiv \frac{\xi_p}{RP_0} = \frac{1}{RP_0} \frac{2g}{\gamma} K(k), \quad (\text{A5})$$

where $K(k)$ is the elliptic integral of the first kind.³⁸

In contrast, when $H = 0$, from Eq. (A3) we have $\eta_2 = \eta_3 = 0$ and there are two eventualities. In the range of values of κ (i.e., for $\kappa \leq -1$ or $\kappa \geq 3$, as shown in Fig. 1) for which the eigensolution $\eta_e = 0$ is stable, $\eta = 0$ is a minimum for the potential $V(\eta)$. In this case the only physical initial condition satisfying $H = 0$ is $\eta_0 = 0$, for which $\eta(\xi) = \eta_0 = 0$. Conversely, whenever $-1 < \kappa < 3$, $\eta = 0$ is an unstable eigensolution ($\eta = 0$ becomes a point of maximum for the potential, as shown in Fig. 6) and infinite initial conditions (η_0, ϕ_0) exist, for which the pump tends to deplete asymptotically (see also Fig. 10). The normalized pump amplitude reads in this case as

$$\eta(\xi) = \frac{\eta_+ \eta_- \{1 - \tanh^2[(\pm\gamma\xi + \delta)/g]\}}{\eta_+ - \eta_- \tanh^2[(\pm\gamma\xi + \delta)/g]}, \quad (\text{A6})$$

where $\eta_+ > \eta_- > \eta(\xi) \geq 0$, with $\eta_+ = 2(3 - \kappa)/7$ and $\eta_- = 2(1 + \kappa)$ for $-1 < \kappa \leq -\frac{1}{2}$, and $\eta_- = 2(3 - \kappa)/7$ and $\eta_+ = 2(1 + \kappa)$ for $-\frac{1}{2} \leq \kappa < 3$, and we set

$$g = \frac{2}{(\eta_+ \eta_-)^{1/2}}, \quad \delta = \int_{\eta_0}^{\eta_-} \frac{d\eta'}{[f(\eta')]^{1/2}}.$$

Further, since the modulus $k^2 = 1$, $K(k)$ tends to infinity and the period diverges [see Eq. (A5) and the solid curves in Figs. 9, 11, and 12].

Finally, whenever the initial conditions are such that $H = H(\eta = 1) = -(\kappa + 1/2)$, from Eq. (A3) we have $\eta_2 = \eta_3 = 1$. As in the previous case, what takes place is either an asymptotic evolution toward the eigensolution $\eta = 1$ or no exchange of power at all, depending on whether the solution $\eta = 1$ is unstable (i.e., for $-4 < \kappa < 0$) or stable

(i.e., for $\kappa \leq -4$ or $\kappa \geq 0$), respectively. If the solution is unstable, the normalized pump power is

$$\eta(\xi) = \frac{\eta_+(1 - \eta_-) - \eta_-(1 - \eta_+)\tanh^2[(\pm\gamma\xi + \delta)/g]}{(1 - \eta_-) - (1 - \eta_+)\tanh^2[(\pm\gamma\xi + \delta)/g]}, \quad (A7)$$

where $1 \geq \eta(\xi) > \eta_+ > \eta_-$, with $\eta_+ = -(1 + 2\kappa)/7$ and $\eta_- = 1 + 2\kappa$ for $-4 < \kappa \leq -1/2$, and $\eta_- = -(1 + 2\kappa)/7$ and $\eta_+ = 1 + 2\kappa$ for $-1/2 \leq \kappa < 0$. Since the modulus $k^2 = 1$, the period diverges [see Eq. (A5) and the dotted line of Fig. 8]. We further set

$$g = \frac{2}{[(1 - \eta_+)(1 - \eta_-)]^{1/2}}, \quad \delta = \int_{\eta_+}^{\eta_0} \frac{d\eta'}{[f(\eta')]^{1/2}}.$$

2. Case II: Different Sideband Amplitudes ($\alpha \neq 0$) with $H = 0$

In the case of different sideband amplitudes we may write the polynomial $f(\eta)$ in the form

$$f(\eta) = \eta^2(\eta - \eta_+)(\eta - \eta_-),$$

where

$$\eta_{\pm} = \frac{1}{2}\{3\kappa + 5 \pm 2[4(\kappa^2 + \kappa + 2) - 7(1 - \alpha^2)]^{1/2}\} \quad (A8)$$

If the value of η is limited between $\eta = 0$ and $\eta = \eta_-$, the solution reduces to the form of Eq. (A6), where the values of η_+ and η_- of Eq. (A3) must be replaced by those of Eq. (A8).

APPENDIX B

This appendix aims at clarifying how the coupled-mode equations governing nondegenerate FWM, whose solutions are reported in Refs. 15 and 16, reduce to the equations governing TWM [see Eqs. (2) above]. We emphasize that the equations governing FWM cannot be correctly reduced to the equations governing the degenerate interaction by means of a simple assumption on the field amplitudes; otherwise one ends up with the wrong coefficients of the nonlinear terms. In fact, the degeneracy implies a subtle limit, which must be performed by properly accounting for all the relevant terms in the polarization expansion, though this has not been correctly recognized in the literature (see, for example, what is stated in Ref. 15).

Four scalar fields vibrating at frequencies ω_1 , ω_2 , ω_3 , and ω_4 [characterized by complex amplitudes E_{ω_i} ($i = 1, 2, 3, 4$)] mix by means of the third-order nonlinearity to yield a polarization

$$P(\omega_1) = 3\chi_{1111}(|E_{\omega_1}|^2 E_{\omega_1} + 2|E_{\omega_2}|^2 E_{\omega_1} + 2|E_{\omega_3}|^2 E_{\omega_1} + 2|E_{\omega_4}|^2 E_{\omega_1} + 2E_{\omega_3}E_{\omega_4}E_{\omega_2}^*), \quad (B1)$$

$$P(\omega_3) = 3\chi_{1111}(|E_{\omega_3}|^2 E_{\omega_3} + 2|E_{\omega_1}|^2 E_{\omega_3} + 2|E_{\omega_2}|^2 E_{\omega_3} + 2|E_{\omega_4}|^2 E_{\omega_3} + 2E_{\omega_1}E_{\omega_2}E_{\omega_4}^*), \quad (B2)$$

where $\omega_1 + \omega_2 = \omega_3 + \omega_4$, and the polarization vibrating at frequency ω_2 may be obtained from Eq. (B1) by interchanging subscripts 1 and 2, and that at ω_4 may be obtained from Eq. (B2) by interchanging subscripts 3 and 4. Equations (B1) and (B2) lead to a self-consistent system of four equations for the complex amplitudes of the fields.^{9,15,16} Following this approach, one implicitly neglects the gen-

eration of several new frequencies. Among these, the ones that are relevant for our analysis are $2\omega_2 - \omega_1$, $\omega_2 + \omega_3 - \omega_1$, and $\omega_2 + \omega_4 - \omega_1$ (as well as those contributions obtained by interchanging subscripts 1 and 2), and $2\omega_1 - \omega_3$ and $2\omega_2 - \omega_3$ (as well as those contributions obtained by interchanging subscripts 3 and 4). These frequencies are associated with products of electric field amplitudes in the polarization expansion, such as $E_{\omega_2}^2 E_{\omega_1}^*$, $2E_{\omega_2}E_{\omega_3}E_{\omega_1}^*$, $2E_{\omega_2}E_{\omega_4}E_{\omega_1}^*$ and $E_{\omega_1}^2 E_{\omega_3}^*$, $E_{\omega_2}^2 E_{\omega_3}^*$.

However, when the process becomes partially degenerate, two photons belong to the same optical field, so that $\omega_3 = \omega_4 = \omega_0$. In this case the above-mentioned contributions to the nonlinear polarization can no longer be neglected, since they now vibrate at one of the frequencies ω_0 , $\omega_1 = \omega_0 - \Omega$, or $\omega_2 = \omega_0 + \Omega$. By retaining these terms in the polarization expansion and by setting $E_{\omega_0} = E_{\omega_3} + E_{\omega_4}$, we obtain for the third-order polarization

$$P(\omega_1) = 3\chi_{1111}(|E_{\omega_1}|^2 E_{\omega_1} + 2|E_{\omega_2}|^2 E_{\omega_1} + 2|E_{\omega_0}|^2 E_{\omega_1} + E_{\omega_0}^2 E_{\omega_2}^*), \quad (B3)$$

$$P(\omega_0) = 3\chi_{1111}(|E_{\omega_0}|^2 E_{\omega_0} + 2|E_{\omega_1}|^2 E_{\omega_0} + 2|E_{\omega_2}|^2 E_{\omega_0} + 2E_{\omega_0}^* E_{\omega_1} E_{\omega_2}); \quad (B4)$$

$P(\omega_2)$ may be obtained from $P(\omega_1)$ by interchanging subscripts 1 and 2. Equations (2) of the main text follow from these expressions for the polarization.

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